

AI Powered Debt Relief & Financial Recovery Platform

Project Description:

The **AI Powered Debt Relief & Financial Recovery Platform** is an intelligent AI-powered web application developed to simplify and automate the debt management and financial recovery process for borrowers. The platform enables users to manage loan details, analyze financial health, generate AI-powered negotiation strategies, and receive adaptive settlement recommendations through a secure and modern digital environment.

Built using **React.js, FastAPI, Python, SQLite, SQLAlchemy ORM, and Google Gemini API**, the system provides a responsive, scalable, and borrower-friendly interface with intelligent financial processing capabilities and real-time AI-assisted decision support.

Scenarios:

Scenario 1: AI-Powered Settlement Recommendation

A borrower adds loan details including outstanding amount, EMI, overdue duration, and monthly income. The system analyzes the financial profile and generates personalized settlement recommendations along with debt stress analysis and financial health insights.

Scenario 2: Intelligent Negotiation Letter Generation

A borrower selects a loan account and requests a negotiation strategy. The platform uses Google Gemini AI to generate professional lender-specific negotiation emails and settlement request letters based on the borrower's financial condition and repayment capability.

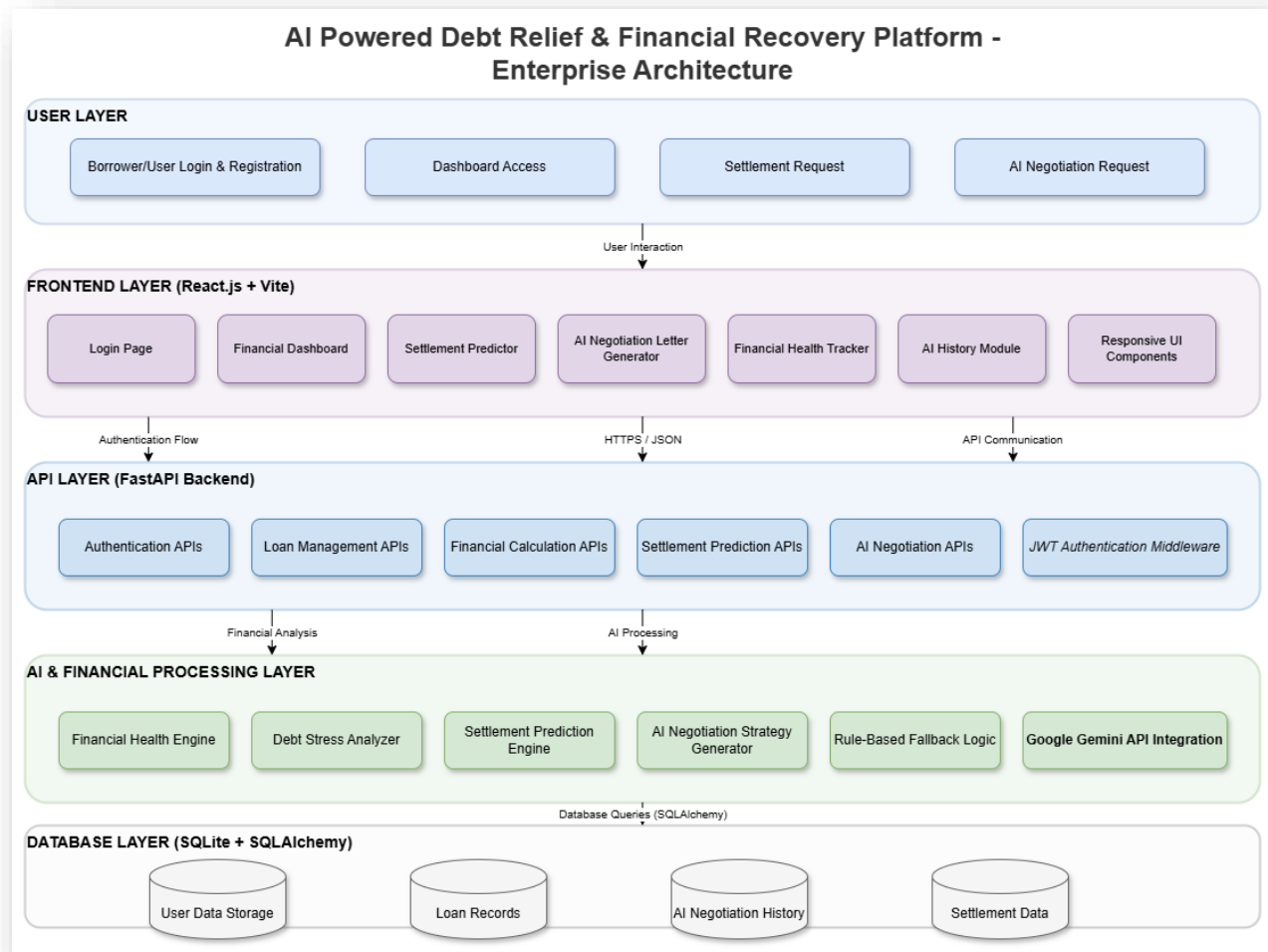
Scenario 3: Financial Health Tracking & Loan Analysis

A borrower accesses the dashboard to monitor financial health metrics such as monthly surplus, EMI ratio, debt stress level, and settlement percentages. The platform also maintains AI-generated negotiation history and provides borrower guidance for improving debt management decisions.

Technical Architecture:

Description: The AI Powered Debt Relief & Financial Recovery Platform follows a scalable 3-tier architecture consisting of a React.js frontend, FastAPI backend, and SQLite database managed using SQLAlchemy ORM. The frontend provides an interactive and responsive user interface for borrowers to manage loans, view financial insights, and access AI-powered recommendations. The backend handles business logic, authentication, financial analysis, settlement calculations, and API communication. The platform integrates Google Gemini API to generate intelligent negotiation strategies, settlement recommendations, and borrower guidance. The database layer securely stores

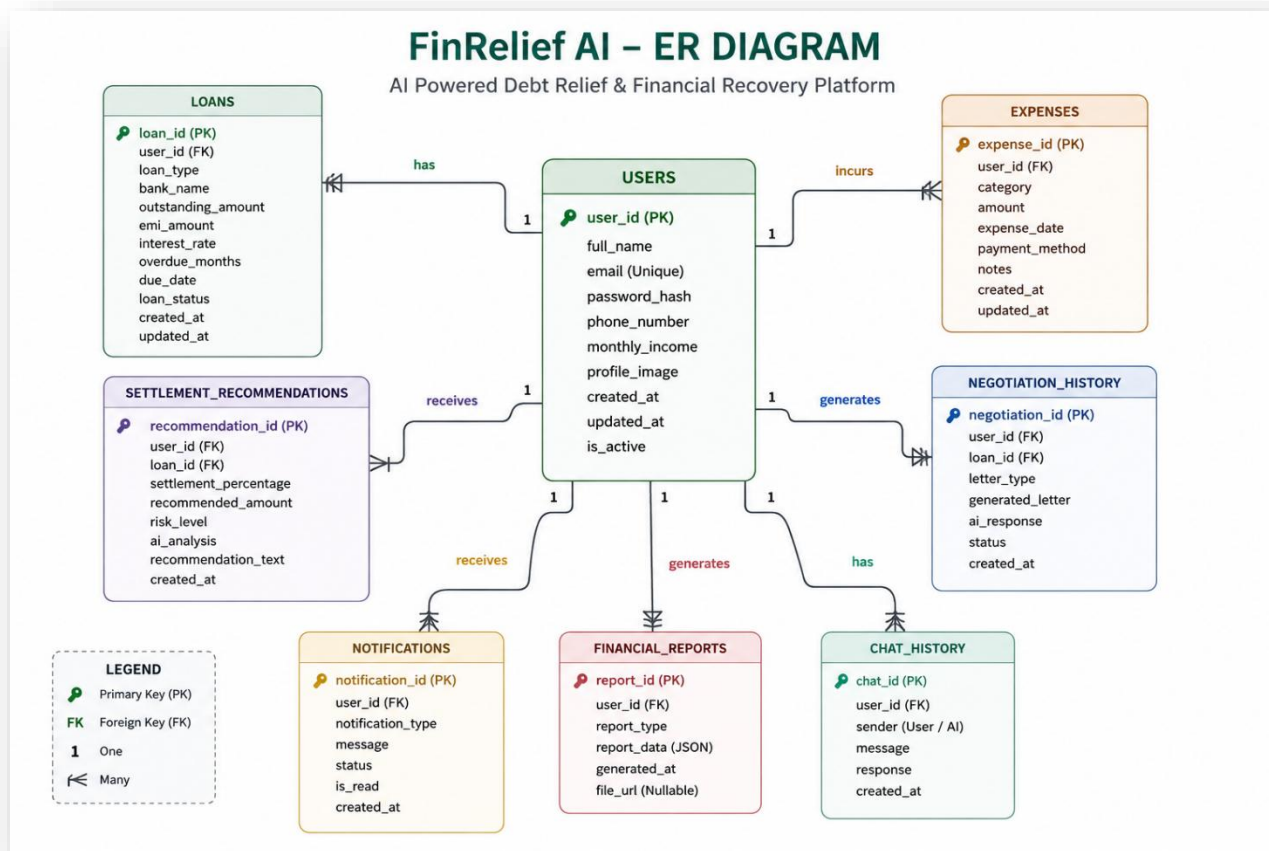
user information, loan records, negotiation history, and financial metrics. This architecture ensures modularity, scalability, security, and efficient processing of borrower financial data with real-time AI-assisted decision support.



ER-Diagram:

Description: The Entity Relationship (ER) Diagram represents the core data structure of the AI Powered Debt Relief & Financial Recovery Platform. The system primarily consists of entities such as Users, Loans, Financial Health Records, Settlement Recommendations, and Negotiation History. A single user can manage multiple loan accounts, where each loan stores information related to outstanding balance, EMI, overdue amount, lender details, and repayment status. The Financial Health module analyzes borrower income, expenses, EMI ratio, and debt stress level to generate insights and recommendations. The Settlement Recommendation entity stores AI-generated settlement plans and repayment suggestions based on borrower financial conditions. The Negotiation History entity maintains records of AI-generated negotiation letters, lender communication strategies, and settlement request history. Relationships between these entities

ensure secure data organization, efficient retrieval, and seamless AI-assisted financial analysis within the platform.



Pre-requisites:

- ❖ **Flask Framework Knowledge:** Understanding of Flask framework, routing, APIs, and backend development concepts.
- ❖ **FastAPI & REST API Basics:** Familiarity with FastAPI architecture, API endpoints, and request handling.
- ❖ **Google Gemini API Familiarity:** Knowledge of integrating AI APIs for generating intelligent financial recommendations and negotiation strategies.
- ❖ **HTML, CSS, and JavaScript Skills:** Basic frontend development skills for designing responsive user interfaces.
- ❖ **React.js Fundamentals:** Understanding of React components, state management, and frontend integration.

- ❖ **Python Programming Proficiency:** Strong understanding of Python programming, libraries, and backend logic implementation.
- ❖ **Database Knowledge:** Familiarity with SQLite database operations and SQLAlchemy ORM usage.
- ❖ **Version Control with Git & GitHub:** Understanding of repository management, commits, branches, and collaboration workflows.
- ❖ **Development Environment Setup:** Installation and configuration of Python, Node.js, VS Code/PyCharm, and required dependencies.
- ❖ **API Testing Knowledge:** Basic experience with Postman or similar tools for API testing and debugging.
- ❖ **Deployment Basics:** Familiarity with deployment tools such as Vercel, Render, or cloud hosting platforms for public application access.

Project Workflow:

Milestone 1: System Planning and Architecture

- **Activity 1.1:** Analyze borrower requirements and identify key debt management challenges.
- **Activity 1.2:** Select suitable AI and backend technologies for financial analysis and settlement recommendation.
- **Activity 1.3:** Design the system architecture defining interactions between frontend, backend, database, and AI services.
- **Activity 1.4:** Set up the development environment and install required dependencies for React.js, FastAPI, SQLite, SQLAlchemy, and Google Gemini API.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop user authentication and secure borrower account management features.
- **Activity 2.2:** Implement loan and debt management modules for storing borrower financial details.
- **Activity 2.3:** Build AI-powered settlement recommendation and debt analysis functionalities.
- **Activity 2.4:** Integrate Google Gemini AI for negotiation strategy and lender communication generation.

Milestone 3: Backend API Development

- **Activity 3.1:** Develop FastAPI backend services for processing borrower data and AI requests.
- **Activity 3.2:** Create REST APIs for loan management, financial analysis, and settlement processing.
- **Activity 3.3:** Configure SQLAlchemy ORM models and database operations using SQLite.

Milestone 4: Frontend Development

- **Activity 4.1:** Design and develop responsive user interfaces using React.js.
- **Activity 4.2:** Implement borrower dashboard, financial health tracking, and loan analysis screens.
- **Activity 4.3:** Integrate frontend with backend APIs for real-time data processing and AI-generated responses.

Milestone 5: Testing and Deployment

- **Activity 5.1:** Perform functional testing, API testing, and performance validation of the platform.
- **Activity 5.2:** Optimize application performance and resolve identified issues.
- **Activity 5.3:** Deploy the application in a secure local or cloud environment for public access.

Milestone 6: Documentation and Conclusion

Milestone 1: System Planning and Architecture

In this milestone, the primary focus is on planning the architecture of the AI Powered Debt Relief & Financial Recovery Platform and selecting the appropriate technologies required for intelligent financial analysis and AI-assisted borrower support. This phase includes setting up the development environment, configuring AI integration, defining the application architecture, and preparing the system for scalable financial data processing.

Activity 1.1: Generate and Configure Google Gemini API Key

Before the platform can communicate with the Google Gemini AI service, a valid API key must be generated and securely configured within the project environment.

Step 1 — Create a Google AI Studio Account

Visit the Google AI Studio platform and sign in using your Google account.

Step 2 — Navigate to API Key Section

After logging in, access the API Keys section from the dashboard to manage AI access credentials.

Step 3 — Create a New API Key

Generate a new API key for the project and assign a descriptive name such as:

finrelief-ai-key

Step 4 — Securely Store the API Key

Copy the generated API key immediately and store it securely because it may not be displayed again later.

Step 5 — Add the API Key to the Project

Create a .env file in the root directory of the project and add the API key as follows:

```
GEMINI_API_KEY=AQ.Ab8RN6LNfgtngQay3zzgypzuw6WiK3Sb3rMD94QosGdDdDGUgA
```

Step 6 — Load the API Key in Backend

The FastAPI backend loads the environment variables using python-dotenv:

```
from dotenv import load_dotenv
import os
```

```
load_dotenv()
```

```
GEMINI_API_KEY = os.getenv("GEMINI_API_KEY")
```

Important Security Note

Never commit the .env file to a public GitHub repository.

Add the following entry inside .gitignore:

```
.env
```

Activity 1.2: Research and Select the Appropriate AI Model

The following research activities were performed to select the most suitable AI model for debt negotiation and financial analysis.

Understand Project Requirements

The platform requires AI capabilities for:

- Debt settlement recommendation

- Financial stress analysis
- Negotiation letter generation
- Borrower financial guidance

Explore Google Gemini Documentation

The available Gemini AI models were analyzed based on:

- Response quality
- Context handling
- Financial reasoning capability
- Text generation performance
- API latency

Evaluate AI Model Performance

Different models were compared based on:

- Accuracy of financial recommendations
- Quality of generated negotiation letters
- Processing speed
- Scalability

Select the Optimal AI Model

The Gemini model was selected because it provides:

- High-quality natural language generation
- Fast response times
- Reliable financial reasoning support

- Scalable API integration capabilities

Activity 1.3: Define the Architecture of the Application

Draft the System Architecture

The platform follows a 3-tier architecture consisting of:

- Frontend Layer
- Backend API Layer
- Database & AI Integration Layer

Frontend Responsibilities

The React.js frontend allows borrowers to:

- Register and log in securely
- Add and manage loan accounts
- Monitor financial health metrics
- Request AI-generated settlement strategies
- View negotiation history and analytics

Backend Responsibilities

The FastAPI backend performs:

- User authentication and authorization
- Loan data processing
- Financial calculations
- Debt stress analysis
- Settlement recommendation generation

- AI request handling

AI Integration Responsibilities

The backend communicates with Google Gemini API to:

- Generate intelligent negotiation letters
- Produce settlement strategies
- Analyze borrower repayment capabilities
- Generate adaptive financial recommendations

Database Layer Responsibilities

The SQLite database with SQLAlchemy ORM stores:

- User accounts
- Loan records
- Financial analysis reports
- AI-generated negotiation history
- Settlement recommendation data

Activity 1.4: Set Up the Development Environment

Install Python and Required Tools

Ensure the following are installed:

- Python 3.10+
- Node.js
- pip package manager
- Git

Create a Virtual Environment

```
python -m venv venv
```

Activate the environment:

```
venv\Scripts\activate
```

Install Backend Dependencies

```
pip install -r requirements.txt
```

Install FastAPI

```
pip install fastapi uvicorn
```

Install SQLAlchemy

```
pip install sqlalchemy
```

Install Google Gemini SDK

```
pip install google-generativeai
```

Install python-dotenv

```
pip install python-dotenv
```

Configure Environment Variables

Create a .env file:

```
GEMINI_API_KEY= AQ.Ab8RN6LNfgtngQay3zzgypzuw6WiK3Sb3rMD94QosGdDdDGUgA
```

Setup Frontend Environment

Install React dependencies:

```
npm install
```

Start the React development server:

npm run dev

Configure Project Structure

The project structure includes:

AI-Debt-Relief-Platform/

- frontend/
- backend/
- database/
- static/
- templates/
- app/
- requirements.txt
- package.json
- .env
- README.md

Milestone 2: Core Functionalities Development

Activity 2.1: Develop Core AI Financial Assistance Features

Implement the core AI-powered financial assistance functionalities including settlement recommendation generation, negotiation strategy creation, and financial health analysis through intelligent AI processing and structured financial data evaluation.

AI Settlement Recommendation Engine

The platform analyzes borrower loan details, income, EMI obligations, overdue duration, and repayment capacity to generate personalized debt settlement recommendations. The AI system evaluates debt stress levels and suggests optimized settlement percentages and repayment strategies.

AI Negotiation Strategy Generator

The platform generates professional lender-specific negotiation letters and settlement request emails using Google Gemini AI. Based on borrower financial conditions and

repayment capability, the system creates personalized negotiation content for improved communication with lenders.

Financial Health Analysis Module

The system calculates and displays financial health indicators such as:

- EMI-to-Income Ratio
- Monthly Surplus
- Debt Stress Level
- Repayment Capacity
- Settlement Feasibility

The platform provides AI-assisted recommendations to help borrowers improve financial stability and debt management.

Activity 2.2: Implement the FastAPI Backend

Define API Routes in FastAPI

Set up API routes in the backend application for:

- User authentication
- Loan management
- Settlement recommendation generation
- Financial health analysis
- AI negotiation strategy generation

Process Borrower Input

Create User Forms in Frontend

Develop forms in React.js for borrowers to:

- Add loan details
- Enter income and EMI information
- Request settlement analysis
- Generate negotiation strategies

Validate User Input

Use backend validation to ensure:

- Required fields are not empty
- Financial values are within valid ranges
- Loan amounts and income values are properly formatted

Handle API Requests

Use FastAPI request handling to process JSON data submitted from the frontend.

Example:

```
from fastapi import FastAPI
```

```
app = FastAPI()
```

Integrate Google Gemini AI API

Generate Structured AI Prompts

The backend dynamically constructs prompts containing:

- Borrower financial profile
- Loan details
- Debt stress indicators
- Settlement requirements

Send Requests to Gemini API

The platform communicates with Google Gemini AI to generate:

- Settlement recommendations
- Negotiation letters
- Financial advice
- Repayment suggestions

Process AI Responses

The backend validates and formats AI-generated responses before returning structured results to the frontend dashboard.

Backend Responsibilities

The FastAPI backend performs:

- Authentication and authorization
- Loan record management
- Financial calculations
- AI communication handling
- Settlement recommendation processing
- API response formatting
- Database interactions using SQLAlchemy ORM

Database Integration

The backend stores:

- User profiles
- Loan information
- AI-generated recommendations

- Negotiation history
- Financial health metrics

using SQLite database with SQLAlchemy ORM for efficient and secure data management.

Milestone 3: Backend Application Development

Milestone 3 focuses on developing and refining the core backend logic of the AI Powered Debt Relief & Financial Recovery Platform. In this phase, the FastAPI backend is configured to handle borrower requests, process financial data, integrate AI-generated responses, and securely communicate with the frontend and database systems.

Activity 3.1: Writing the Main Backend Application Logic

Define the Core API Routes

The backend application defines multiple API routes, each serving a specific functionality within the platform.

Home Route

The root endpoint verifies that the backend service is active and running properly.

```
@app.get("/")
def home():
    return {"message": "AI Debt Relief Platform Backend Running"}
```

Settlement Recommendation Route

This API endpoint accepts borrower loan details and generates AI-powered settlement recommendations.

```
@app.post("/generate-settlement")
def generate_settlement(data: LoanData):
```

The route performs the following operations:

- Receives borrower financial information
- Validates loan and income details
- Calculates debt stress indicators
- Sends structured prompts to Google Gemini AI
- Returns AI-generated settlement recommendations

Negotiation Strategy Route

This endpoint generates intelligent lender negotiation letters using Google Gemini AI.

```
@app.post("/generate-negotiation")  
def generate_negotiation(data: NegotiationRequest):
```

The API:

- Accepts borrower repayment capability and loan information
- Generates professional settlement request letters
- Produces lender-specific negotiation strategies
- Returns formatted AI-generated content

Financial Health Analysis Route

The platform calculates borrower financial metrics through a dedicated API endpoint.

```
@app.post("/financial-analysis")  
def financial_analysis(data: FinancialData):
```

This route computes:

- EMI-to-Income Ratio
- Monthly Surplus
- Debt Stress Level

- Repayment Feasibility
- Financial Risk Indicators

Set Up Route Handlers for Each Feature

Process Borrower Input

The backend processes JSON data submitted from the React.js frontend using FastAPI request models.

Example:

```
from pydantic import BaseModel
```

```
class LoanData(BaseModel):  
    income: float  
    emi: float  
    outstanding_amount: float  
    overdue_months: int
```

Apply Input Validation

The backend validates:

- Empty fields
- Invalid financial values
- Negative loan amounts
- Incorrect borrower information

Validation errors return structured API responses with appropriate HTTP status codes.

Example:

```
if data.income <= 0:  
    return {"error": "Income must be greater than zero"}
```

Define AI Prompt Engineering Helpers

Settlement Prompt Templates

The backend dynamically creates AI prompts based on borrower financial conditions.

Example financial parameters include:

- Monthly income
- EMI obligations
- Outstanding balance
- Overdue duration
- Debt stress level

Financial Risk Categories

The platform categorizes borrowers into:

- Low Risk
- Medium Risk
- High Risk

based on debt burden and repayment capacity.

Build AI Prompt Function

The `build_prompt()` function constructs structured AI prompts for Google Gemini API.

```
def build_prompt(financial_data):  
    return f"""  
    Analyze the borrower financial profile and generate:  
    1. Settlement recommendation  
    2. Debt stress analysis  
    3. Negotiation strategy  
    """
```

Integrate Google Gemini AI Responses

Generate AI Responses

The backend sends borrower financial prompts to Google Gemini API for intelligent processing.

Example:

```
response = model.generate_content(prompt)
```

Process and Validate AI Responses

The system validates AI-generated output before returning results to the frontend dashboard.

The response includes:

- Settlement recommendations
- Negotiation letters
- Financial guidance
- Debt analysis

Database Integration Logic

The backend stores:

- User accounts
- Loan records
- Settlement history
- Negotiation requests
- Financial analysis reports

using SQLite and SQLAlchemy ORM.

Backend Responsibilities Summary

The backend application handles:

- API routing
- Authentication
- Loan data management
- Financial calculations
- AI integration
- Database operations
- Settlement recommendation generation
- Negotiation strategy processing
- Response validation

Core Backend API Endpoints

Endpoint	Method	Purpose
/	GET	Backend health check
/login	POST	User authentication
/register	POST	User registration
/generate-settlement	POST	AI settlement recommendation
/generate-negotiation	POST	AI negotiation strategy
/financial-analysis	POST	Financial health analysis
/loan-history	GET	Retrieve borrower loan history

Milestone 4: Frontend Development

Milestone 4 focuses on designing and developing a modern, responsive, and user-friendly interface for the AI Powered Debt Relief & Financial Recovery Platform. The frontend is built using React.js with responsive UI components that provide borrowers with an intuitive experience for managing loans, viewing financial insights, and accessing AI-powered debt assistance features.

Activity 4.1: Designing and Developing the User Interface

Set Up the Base Frontend Structure

- Develop the main frontend interface using React.js components for modular and reusable UI development.
- Create a responsive dashboard layout including:
 - Navigation Sidebar
 - Borrower Dashboard
 - Loan Management Forms
 - Financial Health Analytics
 - AI Recommendation Sections
 - Negotiation History Panel
- Use semantic HTML structure and accessible UI practices to improve usability and maintainability.
- Integrate modern icons and typography libraries for professional visual appearance.

Design a Responsive Layout Using CSS

- Create a responsive user interface using CSS and modern frontend styling techniques.
- Implement a clean glassmorphism-inspired financial dashboard design for a modern user experience.

- Use Flexbox and Grid layouts to organize dashboard cards, analytics panels, and financial summaries.
- Apply responsive media queries to ensure compatibility across:
 - Desktop
 - Tablet
 - Mobile devices

Create Separate UI Components for Each Feature

Loan Management Cards

- Display borrower loan details including:
 - Outstanding amount
 - EMI
 - Due dates
 - Repayment status
- Provide editable forms for updating loan information.

Financial Health Dashboard

- Display financial metrics such as:
 - EMI-to-Income Ratio
 - Monthly Surplus
 - Debt Stress Level
 - Settlement Percentage
- Present visual indicators and analytical summaries for borrower guidance.

AI Settlement Recommendation Panel

- Display AI-generated settlement suggestions and repayment plans.
- Highlight recommended settlement amounts and repayment feasibility.

AI Negotiation Strategy Section

- Render AI-generated lender negotiation letters inside dedicated UI cards.
- Provide copy-to-clipboard functionality for negotiation content.

Activity 4.2: Creating Dynamic Content Rendering with React.js

Handle Form Submission Asynchronously

- Implement asynchronous API requests using Fetch API or Axios.
- Submit borrower loan and financial details to the FastAPI backend without page refresh.
- Disable submission buttons during processing and display loading animations while AI responses are generated.

Example:

```
const response = await fetch("/generate-settlement", {  
  method: "POST",  
  headers: {  
    "Content-Type": "application/json"  
  },  
  body: JSON.stringify(formData)  
});
```

Render Results Dynamically

- Dynamically render:
 - Settlement recommendations
 - Financial analysis reports

- Debt stress indicators
- Negotiation strategies
- Update dashboard components in real time after receiving backend responses.

Smooth User Navigation

- Automatically scroll users to generated AI results after processing.
- Use smooth transitions and animations for enhanced user interaction.

Restore UI State After Processing

- Re-enable buttons after successful or failed API responses.
- Restore original UI states regardless of processing outcome.
- Display success and error notifications for improved user feedback.

Frontend Responsibilities Summary

The React.js frontend handles:

- User authentication interfaces
- Loan management forms
- Financial dashboards
- AI-generated settlement displays
- Negotiation strategy rendering
- Real-time API interaction
- Responsive mobile-friendly layouts
- User experience enhancements

Milestone 5: Deployment

In Milestone 5, the focus is on deploying the AI Powered Debt Relief & Financial Recovery Platform locally for testing and then hosting it publicly using GitHub and Vercel for real-world accessibility. This phase includes configuring the development environment, verifying application functionality, pushing source code to GitHub, and deploying the frontend application using Vercel.

Activity 5.1: Preparing the Application for Local Deployment

Set Up a Virtual Environment

Open VS Code Terminal or Command Prompt

Create and activate a Python virtual environment:

```
python -m venv venv
```

Activate the environment:

```
venv\Scripts\activate
```

Install all required backend dependencies:

```
pip install -r requirements.txt
```

Configure Environment Variables

Create a .env file in the project root and add the required API keys and configuration values.

```
GEMINI_API_KEY= AQ.Ab8RN6LNfgtngQay3zzgypzuw6WiK3Sb3rMD94QosGdDdDGUGA
```

```
SECRET_KEY=
```

The backend automatically loads these variables using python-dotenv during startup.

Activity 5.2: Local Testing and Verification

Start the Backend Server

Run the FastAPI backend locally:

```
uvicorn main:app --reload
```

The backend runs locally at:

<http://127.0.0.1:8000>

Start the React Frontend

Open another terminal and run:

`npm install`

`npm run dev`

The frontend application runs locally at:

<http://localhost:5173>

Verify Application Features

Test the following modules locally:

- User Registration & Login
- Loan Management
- Financial Health Dashboard
- AI Settlement Recommendation
- AI Negotiation Letter Generation
- Debt Analysis Reports

Ensure API communication, database operations, and AI-generated responses are functioning correctly.

Activity 5.3: Source Code Management Using GitHub

Initialize Git Repository

Open the VS Code terminal and initialize Git:

`git init`

Add Project Files

`git add .`

Commit Source Code

`git commit -m "Initial project commit"`

Connect GitHub Repository

`git remote add origin https://github.com/your-username/AI-Debt-Relief-Platform.git`

Push Code to GitHub

`git branch -M main`

`git push -u origin main`

The complete project source code is now securely hosted on GitHub for version control and collaboration.

Activity 5.4: Public Deployment Using Vercel

Vercel is used to deploy the React.js frontend application publicly with automatic GitHub integration and continuous deployment support.

Connect GitHub Repository to Vercel

- Sign in to Vercel using GitHub account.
- Import the AI-Debt-Relief-Platform repository from GitHub.
- Configure the project settings and environment variables.

Deploy the Application

After configuration, Vercel automatically builds and deploys the application.

The deployed website becomes publicly accessible through a secure Vercel URL such as:

<https://ai-debt-relief-platform.vercel.app>

Deployment Verification

After deployment:

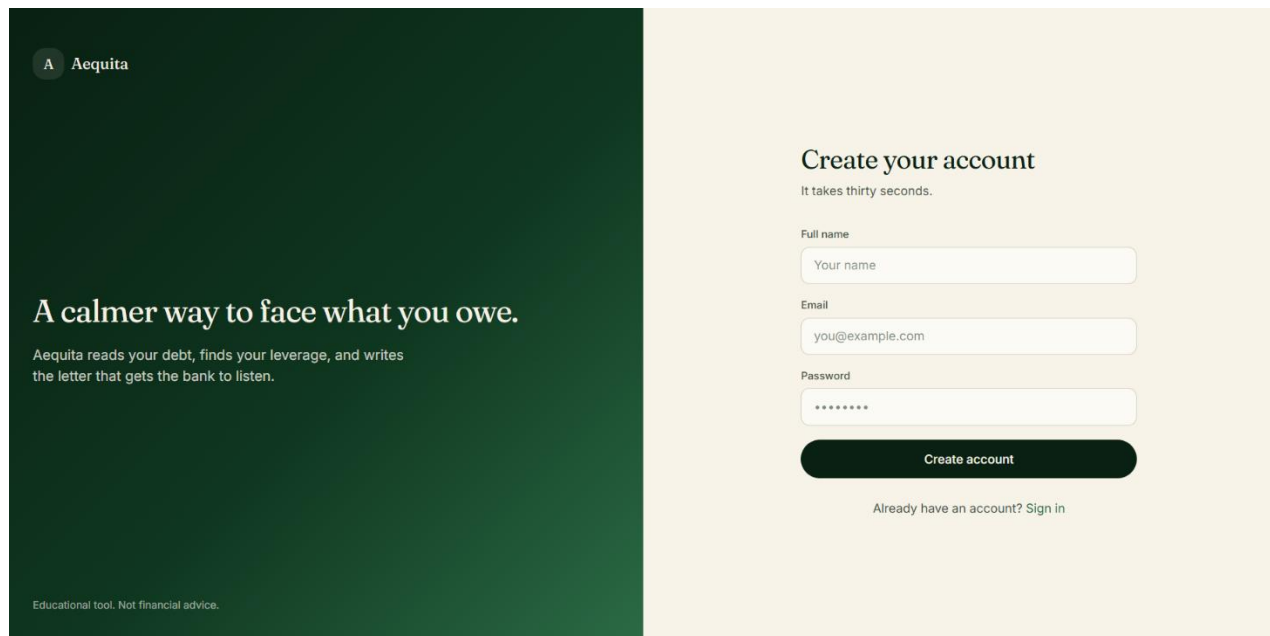
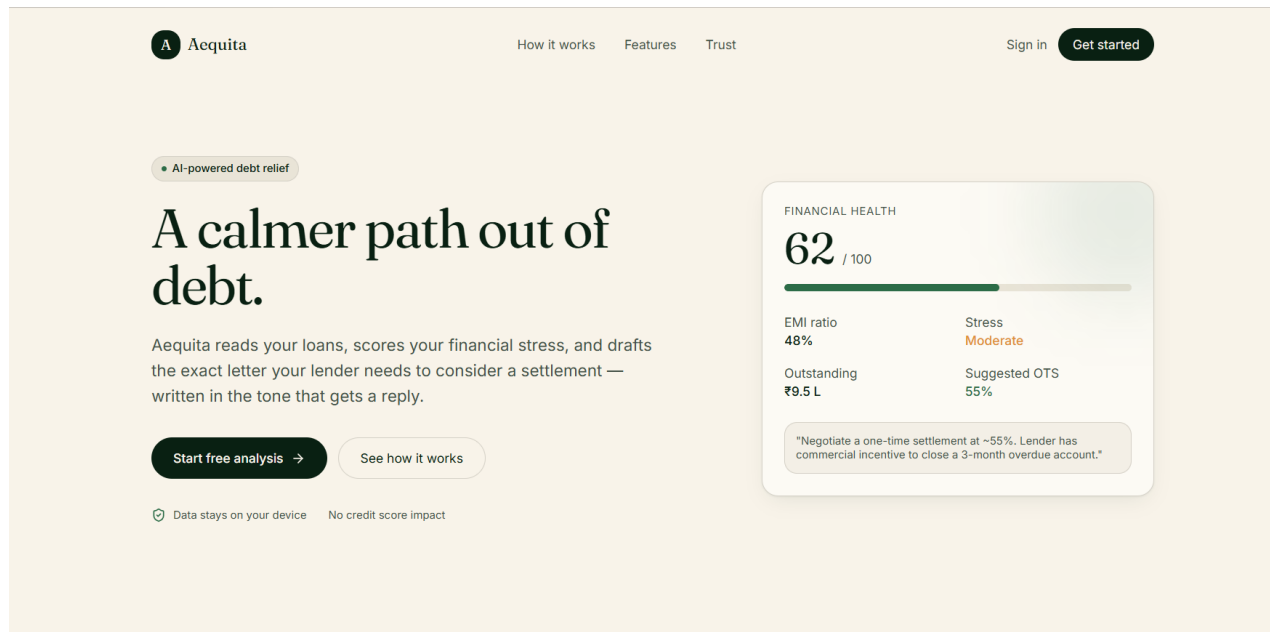
- Verify frontend responsiveness
- Check API integrations
- Test AI-generated settlement recommendations
- Validate negotiation strategy generation
- Confirm database functionality

Important Notes

- Environment variables must be securely configured in Vercel project settings.
- GitHub integration enables automatic deployment whenever new code is pushed.
- Vercel provides fast global CDN delivery and scalable hosting support.
- Backend APIs can be deployed separately using platforms such as Render, Railway, or Azure if required.

Exploring the Website's Web Pages:

Home / Generator Page:



Description: The image represents the homepage of the AI Powered Debt Relief & Financial Recovery Platform named Aequita, designed to help borrowers manage debt through intelligent financial analysis and AI-driven settlement recommendations. The image represents the homepage of the AI Powered Debt Relief & Financial Recovery Platform named Aequita.

Input Section:

Dashboard

Loans

Settlement AI

Negotiation Letter

History

Profile

Kumhari Prasanna
www.prasannakumhari9347@gm...
Sign out

Settlement AI

K

AI-powered settlement engine

Find your leverage, loan by loan.

Aequita blends your overdue history, interest rate, and EMI burden into a realistic one-time settlement percentage you can put to the lender.

PERSONAL LOAN

HDFC Bank

Moderate risk

₹4,50,000 outstanding · 14.5% · 2 mo overdue

Suggested OTS

Settlement offer

60%

₹2,70,000

Negotiate an EMI reduction or interest-rate cut first. Settlement (at ~60%) is a fallback if cashflow worsens.

► Strategy (3 steps)

Draft a settlement letter →

CREDIT CARD

ICICI Bank

₹1,80,000 outstanding · 36% · 3 mo overdue

Run AI analysis

AUTO LOAN

SBI

₹3,20,000 outstanding · 10.25% · current

Run AI analysis

A Aequita

Dashboard

Loans

Settlement AI

Negotiation Letter

History

Profile

Kumhari Prasanna
www.prasannakumhari9347@gm...
Sign out

History

K

Saved AI drafts

Negotiation history

Hardship notice — HDFC Bank

5/7/2026, 2:35:58 pm

👁

🗑

A Aequita

Dashboard

Loans

Settlement AI

Negotiation Letter

History

Profile

Kumhari Prasanna
www.prasannakumhari9347@gmail.com
Sign out

Loans

Your loan portfolio

All loans

+ Add loan

LOAN	OUTSTANDING	EMI	RATE	STATUS	DUE	
Personal Loan HDFC Bank	₹4,50,000	₹14,500	14.5%	2 mo overdue	2026-08-04	
Credit Card ICICI Bank	₹1,80,000	₹9,200	36%	3 mo overdue	2026-08-11	
Auto Loan SBI	₹3,20,000	₹8,900	10.25%	Current	2026-08-19	

A Aequita

Dashboard

Loans

Settlement AI

Negotiation Letter

History

Profile

Negotiation Letter

AI generator

Draft your negotiation letter

Loan account

Personal Loan - HDFC Bank

Letter type

☒ Settlement (OTS)
 One-time settlement offer at a reduced amount.

☐ EMI reduction
 Restructure to lower the monthly EMI.

☐ Hardship notice
 Formal hardship notification, request a meeting.

Settlement offer: 60% of outstanding

Tip: aggressive overdue accounts (3+ months) typically settle at 40-55%.

Your reason / hardship (optional)

e.g. Recent job loss, medical emergency in family, business income reduced by 60%...

Preview

Copy Save PDF Redo

Date: 5 July 2026

To,
The Branch Manager
HDFC Bank

Subject: Request for One-Time Settlement — Personal Loan Account

Respected Sir/Madam,

I, Kumhari Prasanna, am writing with reference to my Personal Loan availed from HDFC Bank, currently carrying an outstanding balance of approximately ₹4,50,000 and a monthly EMI of ₹14,500.

Over the last several months I have been facing a significant reduction in household income combined with rising essential expenses, which has made it impossible to service the current EMI obligations in full. As a consequence, I have been unable to keep my EMI payments fully current, and my account is presently 2 month(s) past due. I sincerely regret any inconvenience caused.

After a careful review of my present and projected cashflow, I am in a position to arrange a one-time settlement amount of ₹2,70,000 (approximately 60% of the outstanding) by liquidating personal savings and assistance from family. I respectfully request the bank to:

Description:

Image 1 – Negotiation History Page

This page shows the borrower's saved AI-generated negotiation letters and previous settlement requests.

Users can view the negotiation history along with the bank name and generated date.

The interface allows users to open or delete saved negotiation drafts easily.

It helps borrowers track all past financial communication in one place.

Image 2 – AI Negotiation Letter Generator

This page allows users to generate professional loan negotiation letters using AI. Users can choose the loan account, settlement type, and provide hardship details. The system automatically creates a formal settlement request letter with financial information.

Users can preview, copy, save, redo, or export the generated letter as a PDF.

Image 3 – Settlement AI Analysis Page

This page displays AI-powered settlement recommendations for different loans.

The system analyzes overdue duration, EMI burden, and financial stress levels.

Users receive suggested settlement percentages and repayment strategies for each loan.

It helps borrowers understand the best negotiation approach for debt recovery.

Image 4 – Loan Management Dashboard

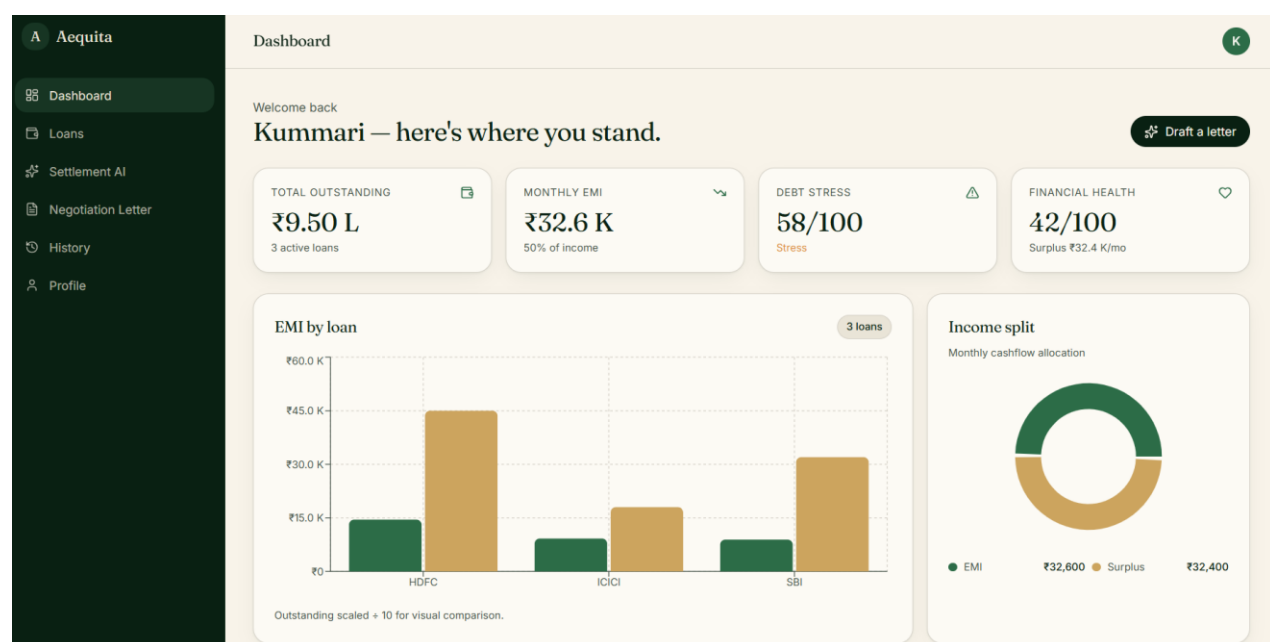
This page provides a complete overview of all borrower loan accounts.

Users can view loan amount, EMI, interest rate, due date, and overdue status.

The dashboard supports adding, editing, and managing multiple loan records easily.

It helps borrowers organize and monitor their financial liabilities effectively.

Results Section:



Description: This dashboard page provides a complete overview of the borrower's financial condition and loan status. It displays important metrics such as total outstanding amount, monthly EMI, debt stress score, and financial health score. The EMI comparison chart visually represents loan-wise payment distribution across different banks. The income split chart helps users understand the balance between EMI expenses and monthly surplus income. The page enables borrowers to monitor their financial recovery progress and make informed debt management decisions.

Conclusion:

The AI-Powered Debt Relief & Financial Recovery Platform is an intelligent web-based financial assistance system developed to simplify and automate debt management and financial recovery processes for borrowers. Built using React.js, FastAPI, Python, SQLite, SQLAlchemy ORM, and Google Gemini AI, the platform provides AI-powered settlement recommendations, financial health analysis, negotiation strategy generation, automated lender communication, and loan management through a secure and user-friendly interface. The system enables borrowers to monitor loan details, analyze debt stress levels, generate professional negotiation letters, and receive personalized settlement guidance in real time. By integrating modern web technologies with artificial intelligence, the project improves financial decision-making, enhances user experience, and reduces manual effort in debt recovery management. The application was successfully deployed using GitHub, VS Code, and Vercel, demonstrating scalable cloud deployment and modern full-stack development practices, while also offering future enhancement opportunities such as real-time banking integration, multilingual support, mobile application support, and advanced predictive financial analytics.